SHIVAM CHAND KAUSHIK

AI/ML Engineer | Semiconductor Domain Expert

Bangalore, India · +91-8310644891 · shivamckaushik@gmail.com · LinkedIn

Professional Summary

AI/ML engineer with 9 years of semiconductor and systems software experience spanning advanced packaging, IC inspection, and factory automation at **Kulicke & Soffa** and **KLA**. Currently pursuing **M.Tech in Artificial Intelligence from IIT Jodhpur**. Building production-grade AI systems—multi-agent architectures, vision-language models, and knowledge graph platforms—specifically for semiconductor manufacturing, inspection, and EDA workflows. One of a rare cohort who combines deep fab-floor domain knowledge (SECS/GEM, wire bonding, pick-and-place, machine vision) with modern AI/ML engineering (LangGraph, PyTorch, LLaVA, RAG, multi-agent orchestration).

Core Competencies

AI/ML & Deep Learning	Semiconductor Domain	Software & Infrastructure
PyTorch, TensorFlow, torchvision	Wafer Defect Inspection (KLA)	C++ (9 years production)
LangGraph, LangChain, RAG	Adv. Packaging & Wire Bonding (K&S)	Python
LLaVA, Qwen-VL, DeepSeek-Coder	SECS/GEM Protocol (Intel integration)	FastAPI
XGBoost, scikit-learn, SHAP	Pick & Place (iFlex, Luminex, Pixalux)	
Computer Vision, CNNs, ViT	SPC, Yield Analysis, Pred. Maintenance	Git/CI-CD
Multi-Agent Orchestration	Smart Factory / Industry 4.0	

Work Experience

Software Engineer — Semiconductor Assembly & Smart Factory AI
Kulicke & Soffa (K&S), Netherlands & Singapore Sept 2020 — Present
via ASM Technologies | K&S is a global leader in semiconductor assembly equipment for advanced packaging.

- Developing and optimizing the C++ software stack for RapidPro and ATPP wire bonding machines, engineering real-time machine control systems that process high-frequency sensor data and manage process parameter optimization.
- Architected product decoupling in the Machine Configuration Tool (MCT), enabling multi-platform support across 6 machine variants (iFlex-T2/T4/H1/CX, Luminex, Pixalux).
- Integrated **MQTT protocol** into MCT for the Smart Factory initiative, enabling real-time machine-to-cloud data streaming for production analytics.
- Led code quality modernization: compiler warning upgrades (W3 to W4), static analysis (PVS-Studio), IWYU implementation, and Boost-to-STL migration.
- Applied AI to K&S semiconductor work: built an **LLM-powered SECS/GEM Protocol Assistant (RAG)** leveraging deep K&S equipment protocol knowledge, and an **ML-based Wire Bond Quality Prediction** system with Bayesian process optimization modeled on RapidPro/ATPP machine parameters—reducing predicted defect rate by 40%.

Software Engineer — IC Inspection & Machine Vision
Quest Global Technologies (client: KLA/ICOS) July 2018 — Sept 2020
Bangalore | KLA is the world's leading semiconductor process control and inspection equipment company.

- Built a Python-based hardware simulator for the ICOS T800 Gantry System—a TCP/IP server modeling machine behavior for offline testing and digital-twin development.
- Developed data logging and analytics pipelines for production metrics: throughput tracking, taper parameter monitoring, and per-head performance analysis.

- Implemented **SECS/GEM interface for Intel**—the semiconductor industry standard protocol for equipment-to-host communication in automated fabs.
- Engineered granular process control: independent per-head yield-loss diagnostics for YZ1/YZ2/YZ4 pick & place heads.

Software Engineer — Telecom & Embedded Systems
Quest Global Technologies (client: Reliance Jio) 2018
Bangalore

- Developed LibWebSocket server for IoT smart speaker integration with ONT router—embedded systems communication architecture for Jio's IoT ecosystem.

Software Engineer — Aerospace & Defense Systems
AxisCades Aerospace & Technology May 2016 — June 2018
Bangalore

- Developed real-time 2D signal visualization for a Bird Detection Radar system using C++/Qt.
- **Led a team of 5 engineers** and commissioned ACRT across **37 units of the Indian Army and Indian Air Force**.

AI/ML Projects

1. Multi-Modal Vision-Language AI for Semiconductor Inspection & Autonomous FA Report Generation GitHub | HuggingFace Dataset | arXiv:2604.13236

- Built a four-agent LangGraph pipeline (Defect Describer, Root Cause Analyzer, Severity Classifier, Recipe Advisor) that fuses DINOv2 visual embeddings, SECS/GEM-format equipment telemetry, and a Qdrant historical defect retrieval database into a unified LLaVA-1.6 context—generating structured Failure Analysis reports in 48 seconds, replacing a 2–4 hour manual process.
- Trained a DINOv2 + MLP defect classifier on SemiFA-930, a self-constructed 930-image dataset spanning 9 semiconductor defect classes (SEM, optical, wafer map); achieved 92.1% accuracy / 0.917 Macro F1 with only 214K trainable parameters vs. 23.5M for ResNet-50 (82.9%).
- Publicly released SemiFA-930 dataset on HuggingFace; paper published on arXiv (cs.CV) — arXiv:2604.13236.
- **Stack:** LLaVA-1.6, DINOv2, QLoRA/PEFT, LangGraph, SECS/GEM, Qdrant, FastAPI, ReportLab, Streamlit

2. Wafer Defect Detection & Classification (WM-811K) GitHub

- Built an end-to-end deep learning pipeline for wafer defect pattern classification on the WM-811K dataset (811K+ wafer maps, 9 defect classes) using ResNet18 and EfficientNet-B0, with a ConvAutoEncoder for anomaly detection on unseen patterns. Achieved 97.83% overall test accuracy and 91.95% defect-class accuracy on a severely class-imbalanced dataset (103,201 normal vs 104 Near-Full samples). Improved Scratch F1 from 0.48 → 0.820 by moving from 64×64 to 128×128 resolution, adding FocalLoss(γ=2) and synthetic scratch augmentation; diagnosed critical training bugs across normalization, BatchNorm domain shift, loss double-suppression, and `torch.compile` checkpoint corruption.
- Trained a two-stage classifier (binary Stage 1 → 8-class Stage 2) for comparison: Stage 1 achieved 98.45% binary accuracy but error propagation reduced combined accuracy to 96.74%, confirming single-stage as the correct production model. Applied 8-view Test-Time Augmentation (+0.33% overall, stabilises low-support classes) and built a v2+v3 ensemble classifier with per-class confidence thresholding and human-review routing for safety-critical predictions.
- **Stack:** PyTorch, torchvision, scikit-learn, FastAPI

3. LLM-Powered SECS/GEM Protocol Assistant (RAG) *(Applied at Kulicke & Soffa)*

- Built a RAG-powered LLM assistant for the SECS/GEM semiconductor protocol, enabling natural language queries over SEMI standards, equipment specifications, and alarm codes. Fine-tuned retrieval on SECS/GEM documentation with FAISS vector store.
- Deployed as an interactive web application for equipment and process engineers in semiconductor fabs.
- **Stack:** HuggingFace Transformers, LangChain, FAISS, Streamlit

4. Wire Bond Quality Prediction & Process Optimization *(Applied at Kulicke & Soffa)*

- Developed ML-based wire bond quality prediction using process data (bond force, ultrasonic power, temperature, loop height) modeled on K&S RapidPro and ATPP machine parameters. Classified bond pass/fail with root cause analysis and applied Bayesian optimization to identify optimal bonding parameters, reducing predicted defect rate by 40%.
- **Stack:** scikit-learn, Optuna, matplotlib

Education

Indian Institute of Technology (IIT), Jodhpur June 2024 — May 2026
Master of Technology (M.Tech) — Artificial Intelligence

JSS Academy of Technical Education, Noida June 2011 — May 2015
Bachelor of Technology (B.Tech) — Information Technology

Achievements

- **Bright Mind Award** — ASM Technologies Ltd. (2021), recognizing exceptional technical contribution in semiconductor software engineering
- **779th Rank** — International Mathematics Olympiad (IMO), demonstrating strong analytical and quantitative aptitude

Certifications (In Progress)

Certification	**Provider**
Fundamentals of Deep Learning	NVIDIA Deep Learning Institute (DLI)
AWS Certified Machine Learning Engineer — Associate	Amazon Web Services

Project repositories available on GitHub upon request.